

W15e - Isotherms of Real Gases

Group 6: Laiba Rehan Zahid - 3771917 (50%), Syed Muhammad Haider Amir - 3787940 (50%)

December 11, 2025

1 Introduction

Real gases deviate significantly from ideal behaviour when they are compressed to high pressures or cooled close to their condensation point. In this regime, intermolecular forces and the finite size of the molecules can no longer be neglected, and simple models such as the ideal gas law

$$pV = nRT \quad (1)$$

fail to describe the observed pressure–volume–temperature (p – V – T) relations. A quantitative understanding of these deviations is essential for thermal physics and for many practical applications in chemistry and engineering.

A first refinement of the ideal gas model is provided by the van der Waals equation of state, which introduces two empirical parameters a and b to account for attractive intermolecular forces and the finite molecular volume, respectively:

$$\left(p + \frac{a}{V_m^2}\right) (V_m - b) = RT. \quad (2)$$

Here $V_m = V/n$ is the molar volume. In the p – V diagram (Clausius–Clapeyron diagram), the van der Waals isotherms reproduce several characteristic features of real gases, such as the emergence of an S-shaped region at temperatures below the critical temperature T_k . In this region, a horizontal segment, the Maxwell construction, represents the coexistence of liquid and vapour at the saturation vapour pressure $p_s(T)$. At the critical point (p_k, V_k, T_k) , the distinction between liquid and gas disappears and the isotherm has an inflection point.

A central quantity associated with the liquid–vapour phase transition is the molar latent heat of vaporization Q_{23} , which relates the energy required to convert one mole of liquid into vapour at constant pressure and temperature. The temperature dependence of the saturation vapour pressure is governed by the Clausius–Clapeyron equation

$$\frac{dp_s}{dT} = \frac{Q_{23}}{T \Delta V_m}, \quad (3)$$

where $\Delta V_m = V_{m,G} - V_{m,L}$ is the difference between the molar volumes of the gaseous and liquid phases. Under the additional assumptions that the vapour behaves approximately as an ideal gas and that Q_{23} is temperature independent over a limited range, Eq. (3) integrates to the familiar exponential vapour–pressure law

$$p_s(T) = p_{s0} \exp\left(-\frac{Q_{23}}{RT}\right), \quad (4)$$

which predicts a linear relationship between $\ln p_s$ and $1/T$.

In this experiment, we investigate the isotherms of sulfur hexafluoride (SF_6) in a high–pressure glass capillary at several fixed temperatures. From the measured p – V data we construct isotherms, identify the Maxwell plateau and determine the saturation vapour pressure $p_s(T)$ for each temperature. A linear fit of $\ln p_s$ versus $1/T$ yields an effective average value of the molar latent heat of vaporization. The amount of substance n contained in the capillary is obtained by extrapolating the product pV to the ideal–gas limit at large volumes. Finally, using the Clausius–Clapeyron equation in differential form, we determine the temperature dependence $Q_{23}(T)$ and analyse the behaviour of the latent heat as the critical point is approached.

Beyond providing numerical values for Q_{23} , n and the critical parameters, this experiment offers a direct visualisation of phase coexistence, the disappearance of the liquid–vapour interface, and critical opalescence near the critical point. A careful treatment of experimental uncertainties in pressure, volume and temperature, as well as in the various fits and extrapolations, is essential to assess the reliability of the extracted thermodynamic quantities and to compare them with theoretical predictions and literature data.

2 Theoretical Analysis

The behaviour of real gases deviates markedly from ideality when the temperature approaches the condensation region or when strong compression is applied. In such regimes, intermolecular interactions and finite molecular volume influence the equation of state and modify the structure of isotherms in the p - V diagram. The following subsections provide a detailed theoretical framework for analysing the isotherms of SF_6 , determining the saturation vapour pressure, extracting the molar latent heat of vaporization, evaluating its temperature dependence, and determining the amount of substance contained in the measurement cell. All formulas required for the later data analysis are derived here.

2.1 Ideal Gas Model and Equation of State

For an ideal gas consisting of n moles at temperature T , pressure p , and volume V , the equation of state is

$$pV = nRT, \quad (5)$$

where R is the molar gas constant. Introducing the molar volume $V_m = V/n$ yields the one-mole form

$$pV_m = RT. \quad (6)$$

The ideal gas model neglects both intermolecular forces and the finite molecular size. It is therefore accurate only at low pressures and sufficiently high temperatures. In the vicinity of liquefaction or near the critical point, corrections to Eq. (5) become essential.

2.2 Real Gas Behaviour and the van der Waals Equation

A simple refinement that incorporates molecular attractions and finite molecular volume is the van der Waals equation,

$$\left(p + \frac{a}{V_m^2}\right)(V_m - b) = RT, \quad (7)$$

where

- a quantifies attractive (cohesive) intermolecular forces,
- b represents the excluded volume per mole.

Eq. (7) is a cubic equation in V_m for fixed p and T , capable of predicting non-monotonic isotherms with an S-shaped region indicating metastable and unstable states. Below the critical temperature, this region is replaced physically by a horizontal line (Maxwell construction) representing liquid-vapour coexistence.

2.2.1 Critical Point Derivation

To obtain the critical point (p_k, V_k, T_k) , we use the fact that at the critical point the isotherm has an inflection point:

$$\left(\frac{\partial p}{\partial V_m}\right)_{T_k} = 0, \quad \left(\frac{\partial^2 p}{\partial V_m^2}\right)_{T_k} = 0. \quad (8)$$

Solving these two conditions using Eq. (7) yields the well-known relations

$$V_k = 3b, \quad T_k = \frac{8a}{27bR}, \quad p_k = \frac{a}{27b^2}. \quad (9)$$

These expressions allow an estimate of the critical parameters of SF_6 using its tabulated a and b constants.

2.3 Isotherms and the Maxwell Construction

Below the critical temperature, van der Waals isotherms exhibit a region where

$$\frac{\partial p}{\partial V_m} > 0,$$

which is thermodynamically unstable. In a real system, this region is replaced by a flat portion at constant pressure $p_s(T)$, the saturation vapour pressure. This pressure is determined experimentally by measuring the plateau of the isotherm, corresponding to coexistence between liquid and vapour. The area-equal rule (Maxwell construction) ensures equal Gibbs free energies of the two phases.

2.4 Clausius–Clapeyron Equation and Latent Heat of Vaporization

The fundamental relation governing the slope of the liquid–vapour coexistence curve is the Clausius–Clapeyron equation:

$$\frac{dp_s}{dT} = \frac{Q_{23}}{T \Delta V_m}, \quad (10)$$

where

- Q_{23} is the molar latent heat of vaporization,
- $\Delta V_m = V_{m,G} - V_{m,F}$ is the difference between the molar volumes of gas and liquid.

2.4.1 Derivation of the Clausius–Clapeyron Equation

Starting from the Gibbs–Duhem relation for one phase,

$$SdT - Vdp + n d\mu = 0, \quad (11)$$

and applying it to the liquid and vapour phases separately, subtracting gives

$$(S_G - S_F)dT - (V_G - V_F)dp_s = 0. \quad (12)$$

At phase coexistence, $\mu_G = \mu_F$, and the molar entropy difference equals

$$\Delta S_m = \frac{Q_{23}}{T}.$$

Substituting yields Eq. (10).

2.4.2 Integrated Form for Low Temperatures

If the vapour behaves ideally and the liquid volume is negligible,

$$V_{m,G} \approx \frac{RT}{p_s}, \quad V_{m,F} \ll V_{m,G},$$

then from Eq. (10),

$$\frac{dp_s}{dT} \approx \frac{Q_{23}p_s}{RT^2}.$$

Rearranging gives

$$\frac{1}{p_s} \frac{dp_s}{dT} = \frac{Q_{23}}{RT^2}. \quad (13)$$

Integrating between T_0 and T :

$$\ln\left(\frac{p_s(T)}{p_s(T_0)}\right) = -\frac{Q_{23}}{R} \left(\frac{1}{T} - \frac{1}{T_0}\right).$$

Thus the experimentally useful form is

$$p_s(T) = p_{s0} \exp\left(-\frac{Q_{23}}{RT}\right), \quad (14)$$

which predicts a linear relation between $\ln p_s$ and $1/T$. The slope directly yields Q_{23} .

2.5 Temperature Dependence of the Latent Heat

Near the critical point, both $Q_{23}(T)$ and $\Delta V_m(T)$ become strongly temperature-dependent. Therefore Eq. (14) is not reliable for extracting the true latent heat close to T_k . Instead, one must use Eq. (10) directly:

$$Q_{23}(T) = T \Delta V_m(T) \left(\frac{dp_s}{dT}\right). \quad (15)$$

Here,

- dp_s/dT is obtained from a smooth fit to the measured $p_s(T)$ values,
- $\Delta V_m(T)$ is extracted from the isotherms using molar volume values derived from the amount of substance.

This provides a temperature-resolved determination of the latent heat and allows one to study its critical behaviour as $T \rightarrow T_k^-$.

2.6 Determination of the Amount of Substance

At sufficiently large volumes (low densities) the gas approaches ideal-gas behaviour. Taking the limit $V \rightarrow \infty$ in Eq. (5) yields

$$\lim_{V \rightarrow \infty} pV = nRT. \quad (16)$$

Experimentally, one plots the product pV as a function of $1/V$. For a real gas, this relation can be approximated by a linear function near the ideal region:

$$pV = A + B \left(\frac{1}{V} \right). \quad (17)$$

Extrapolating to $1/V = 0$ gives $A = nRT$, from which the amount of substance follows:

$$n = \frac{A}{RT}. \quad (18)$$

This value is required for converting volumes into molar volumes and for evaluating ΔV_m in Eq. (15).

3 Experimental Analysis

This section describes the instrumentation used in the experiment and outlines the complete measurement procedure required to obtain pressure–volume isotherms of SF_6 at eight different temperatures. Emphasis is placed on the correct handling of the high–pressure apparatus, adherence to safety limits, and ensuring thermal and mechanical equilibrium during data acquisition.

3.1 Experimental Setup

The experiment is carried out using a commercial apparatus designed for the investigation of the critical point of gases. Its main components are a thick-walled high-pressure glass capillary containing SF_6 , a mercury-based compression system, a pressure gauge, and a temperature-controlled water bath.

Glass Capillary and Volume Scale. The SF_6 sample is sealed inside a vertical glass capillary with a calibrated millimetre scale attached to its outer surface. This allows direct reading of the gas volume. The capillary is enclosed within a transparent Plexiglas water jacket connected to an external thermostat, ensuring uniform temperature control.

Mercury Compression System. Pressure is applied to the gas by raising a mercury column from below the capillary using a handwheel-driven piston mechanism. Turning the handwheel slowly increases the pressure transmitted to the enclosed gas. The use of mercury guarantees nearly incompressible pressure transfer, but also introduces strict safety requirements.

Manometer. A high-precision analogue manometer is connected directly to the mercury chamber and provides pressure readings up to 50 bar. The manometer resolution constitutes one of the limiting factors in pressure measurement accuracy.

Temperature Regulation. A circulating thermostat heats or cools a water bath surrounding the capillary. A digital thermometer immersed in the bath measures the temperature with high stability. The system must be allowed to reach thermal equilibrium before any volume or pressure readings are taken.

Safety Considerations. Due to the presence of mercury and high internal pressures, the following safety limits must be strictly observed:

- The pressure must *never* exceed 50 bar.
- The temperature of the thermostat bath must remain below 55 °C.
- Temperature adjustments may not be performed while the mercury column is raised.

These constraints prevent capillary rupture and mercury release. All measurements are performed slowly to avoid abrupt changes in internal pressure and to ensure thermal



Figure 1: Experimental Setup

3.2 Data Acquisition and Measurement Procedure

The experiment consists of recording eight isotherms of SF_6 at the following temperatures:

$$T = \{\text{room temperature}, 33^\circ\text{C}, 36^\circ\text{C}, 39^\circ\text{C}, 42^\circ\text{C}, 45^\circ\text{C}, 47^\circ\text{C}, 50^\circ\text{C}\}.$$

Each isotherm requires measuring pressure as a function of volume while maintaining a constant temperature.

1. Temperature Stabilisation. The thermostat is set to the desired temperature. Sufficient time must be allowed for the water bath, the capillary, and the enclosed gas to reach thermal equilibrium. The temperature reading must remain stable before a measurement sequence begins.

2. Adjustment of Pressure and Volume. Pressure is increased gradually by turning the handwheel, which raises the mercury column. After each adjustment:

- wait until the pressure reading on the manometer stabilises,
- ensure no visible oscillation or drift of the mercury–gas boundary,
- then read off the volume from the capillary scale.

Each data point consists of a pair (p, V) . At least ten points are required per isotherm to map the full behaviour of the gas.

3. Recording the Coexistence Region. As the pressure increases at temperatures below the critical temperature, the system eventually enters the liquid–vapour coexistence region. In this portion of the isotherm, the pressure no longer increases significantly with reduced volume. Two or more data points from this nearly horizontal region must be recorded. These points allow the determination of the saturation vapour pressure $p_s(T)$.

4. Measurements Near the Critical Point. Around 50°C , the isotherms become increasingly flat and the distinction between liquid and vapour phases diminishes. In addition to quantitative measurements, qualitative observations are recorded:

- the disappearance of the meniscus,
- the formation of fog or opalescence,
- the difficulty in identifying a phase boundary.

These features provide visual evidence of critical phenomena.

5. Highest-Temperature Isotherm for Determining n . The isotherm at the highest temperature (50°C) shows the smallest deviation from ideal behaviour and is therefore used to determine the amount of substance via extrapolation of pV to the ideal-gas limit. Multiple low-pressure points (large volume) are taken to guarantee reliable linear regression.

6. Repetition for All Temperatures. The above steps are repeated for every listed temperature. Care must be taken to avoid overshooting the pressure or temperature limits. After finishing an isotherm, the system must be depressurised and allowed to relax before adjusting the temperature to the next setting.

4 Determination of the Saturation Vapour Pressure from Isotherms (Task 2)

4.1 Goal

The aim of this task is to analyse the measured isotherms $p(V)$ of SF_6 at eight different temperatures and to determine, for each isotherm, the saturation vapour pressure $p_s(T)$. In the p - V diagram of a real gas below the critical temperature, the liquid–vapour coexistence region appears as a nearly horizontal segment (Maxwell plateau). The pressure along this plateau corresponds to the saturation vapour pressure. Extracting these values provides the experimental basis for the subsequent vapour-pressure analysis and the determination of the latent heat of vaporization.

4.2 Theoretical Background

For temperatures below the critical temperature T_k , real-gas isotherms exhibit a region of phase coexistence where the pressure remains approximately constant as the volume decreases. This horizontal portion of the isotherm is described by the Maxwell construction. The pressure in this region is the *saturation vapour pressure* $p_s(T)$.

Mathematically, the saturation pressure is the pressure at which the molar Gibbs free energies of the gas and liquid phases are equal:

$$G_{\text{gas}}(T, p_s) = G_{\text{liq}}(T, p_s). \quad (19)$$

Experimentally, it is obtained from the flattest part of the measured isotherm. The resulting values will later be used in the Clausius–Clapeyron analysis:

$$\frac{dp_s}{dT} = \frac{Q_{23}}{T(V_{m,G} - V_{m,F})}, \quad (20)$$

and in the integrated vapour-pressure law,

$$\ln p_s = \ln p_{s0} - \frac{Q_{23}}{RT}, \quad (21)$$

which is valid sufficiently far from the critical region.

4.3 Method of Analysis

Each isotherm was plotted as pressure versus volume using the measured values from the capillary scale (in mm) and manometer (in bar). The experimental uncertainties of the instruments were taken as:

$$u(p) = \pm 0.25 \text{ bar}, \quad (22)$$

$$u(V) = \pm 0.05 \text{ mm}. \quad (23)$$

To determine the saturation pressure, the flattest three-point window in each isotherm was identified. For each temperature T , the saturation pressure was defined as the mean value of the pressure within this plateau region:

$$p_s(T) = \frac{1}{N} \sum_{i=1}^N p_i, \quad (24)$$

with $N = 3$. The statistical uncertainty of the mean was computed as

$$u_{\text{stat}}(p_s) = \frac{\sigma_p}{\sqrt{N}}, \quad (25)$$

and combined with the instrumental pressure uncertainty using

$$u(p_s) = \sqrt{u(p)^2 + u_{\text{stat}}(p_s)^2}. \quad (26)$$

All calculations were performed using a Python script in a Jupyter notebook. Error bars were included in the p - V plots, and the resulting saturation pressure for each temperature was indicated by a red dashed horizontal line.

4.4 Results

Figures 2–9 show the measured isotherms at the eight temperatures, including uncertainties in both axes. The extracted saturation vapour pressures are summarised in Table 1.

Table 1: Saturation vapour pressures $p_s(T)$ obtained from the Maxwell plateau of each isotherm.

Set Temperature T (°C)	p_s (bar)	$u(p_s)$ (bar)
20	15.45	0.27
33	28.17	0.30
36	30.00	0.25
39	32.00	0.25
42	34.67	0.30
45	37.00	0.25
47	38.67	0.51
50	40.58	0.44

4.5 Plots

Below are the isotherm plots used to extract the saturation pressures.

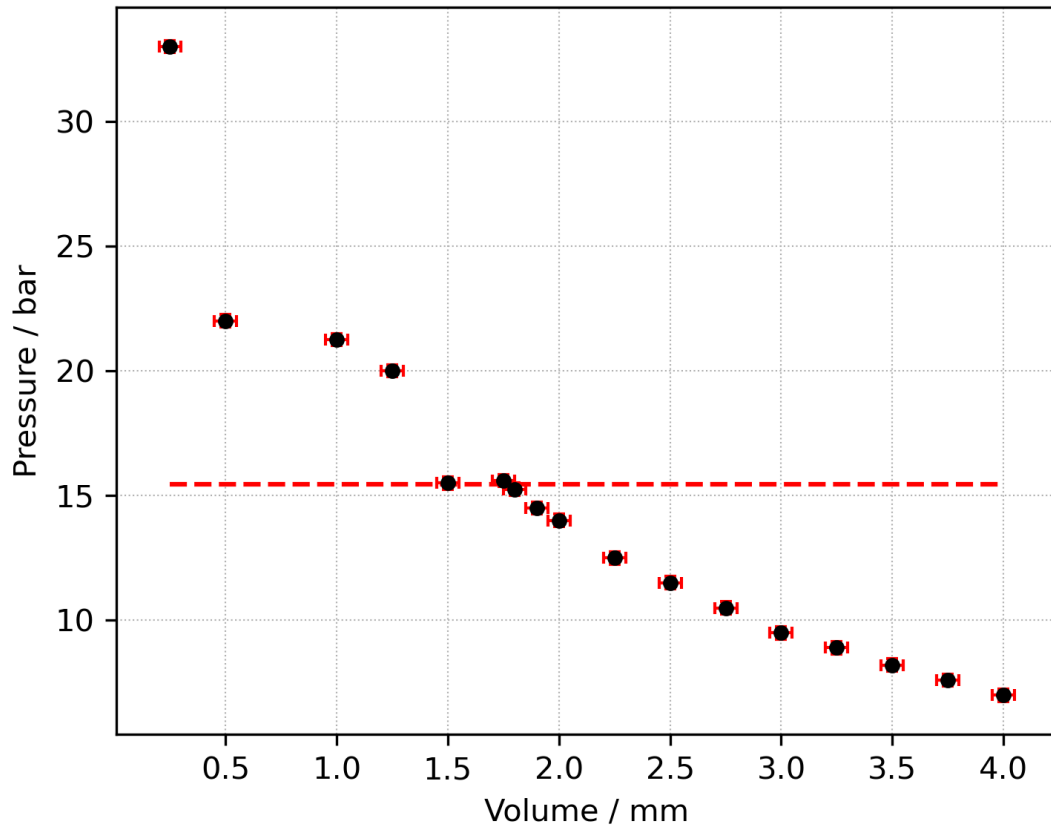


Figure 2: Isotherm at 20°C with saturation pressure indicated by the red dashed line.

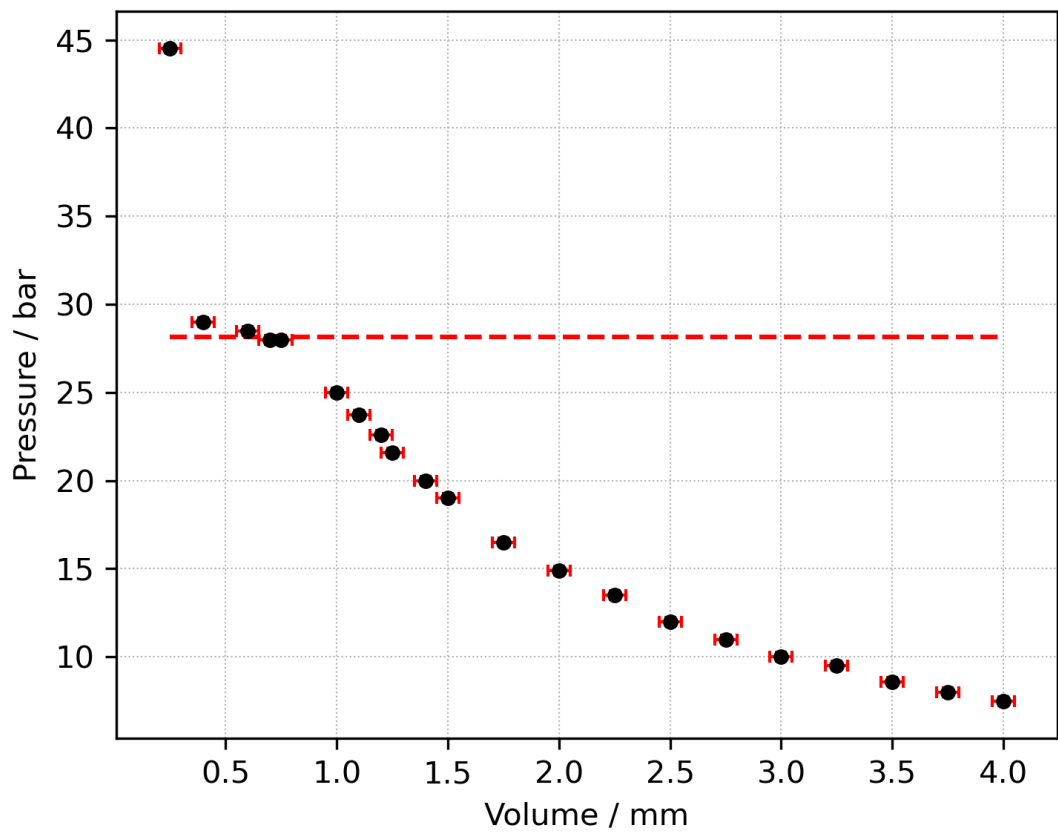


Figure 3: Isotherm at 33°C.

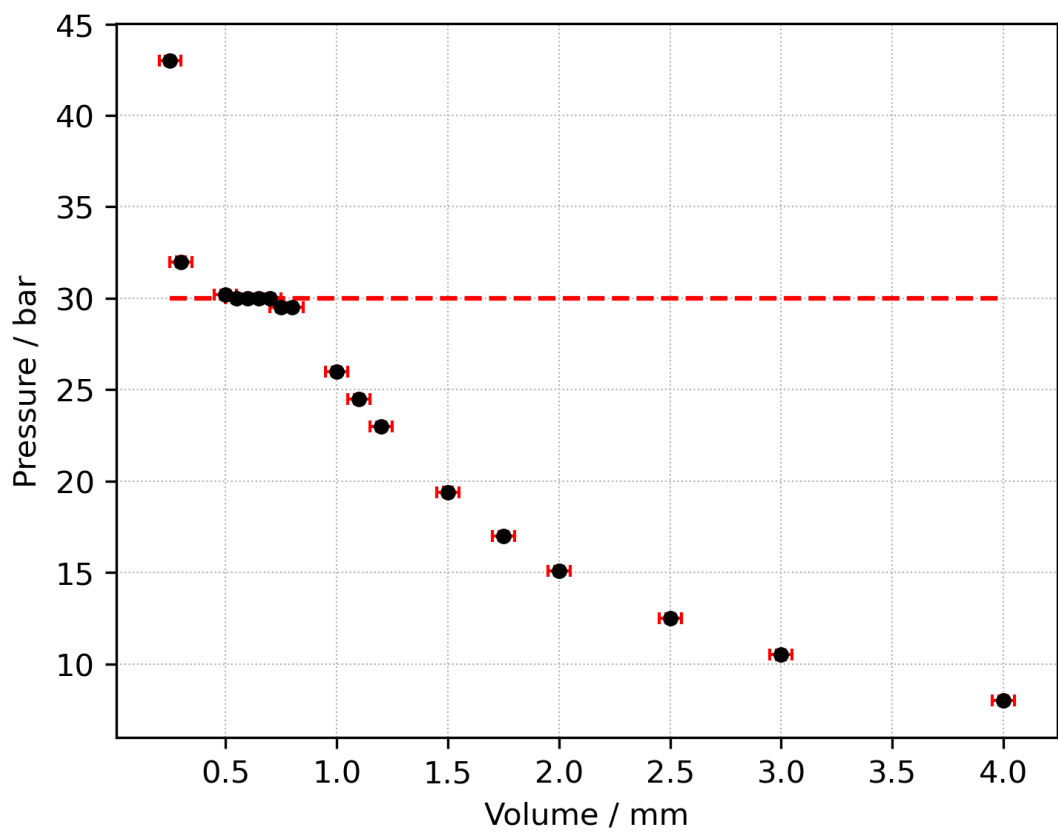


Figure 4: Isotherm at 36°C.

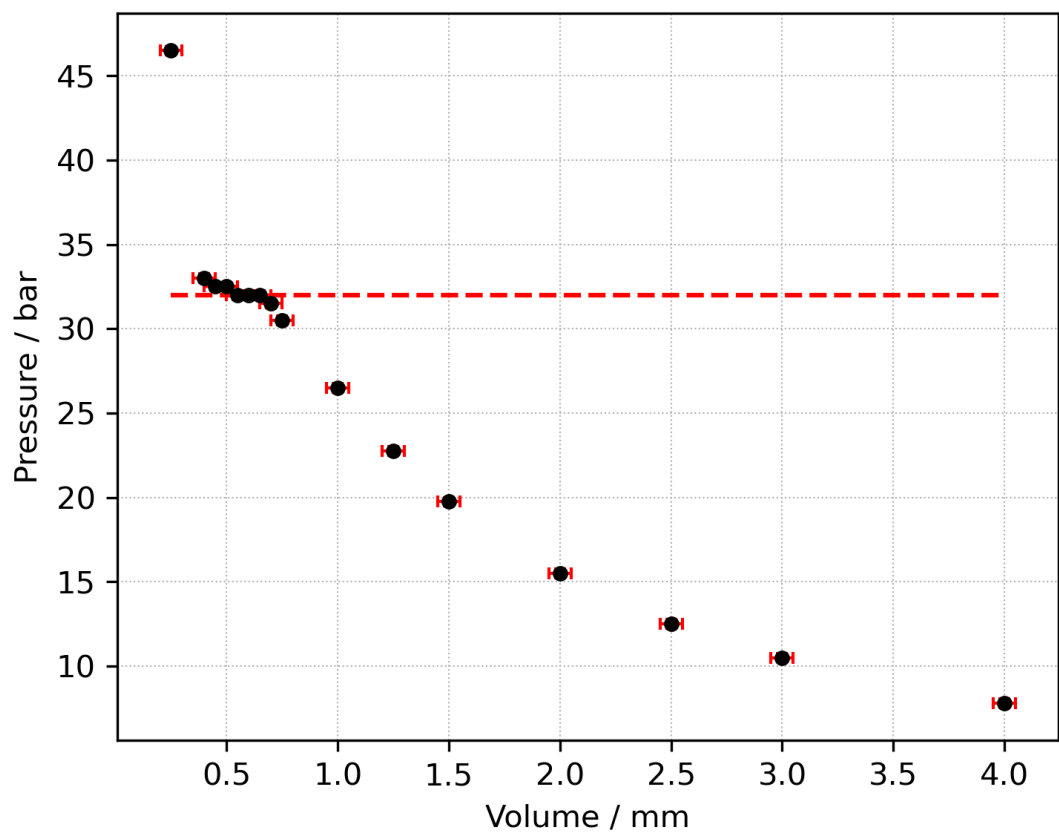


Figure 5: Isotherm at 39°C.

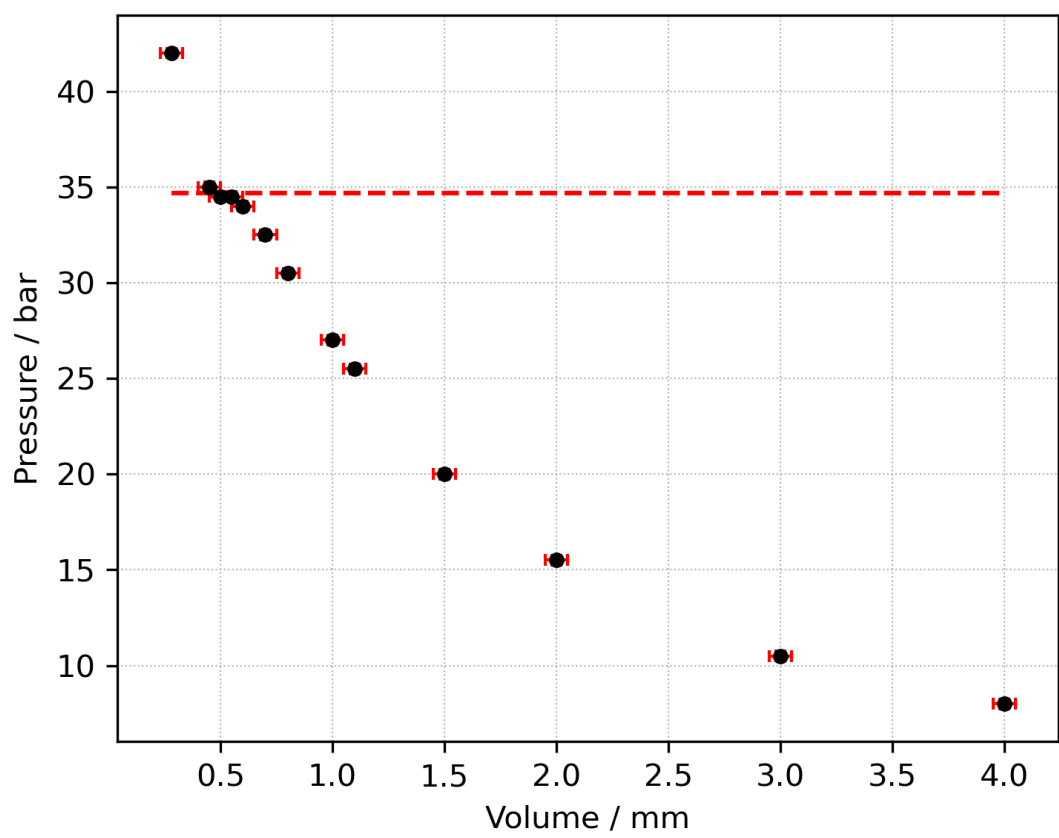


Figure 6: Isotherm at 42°C.

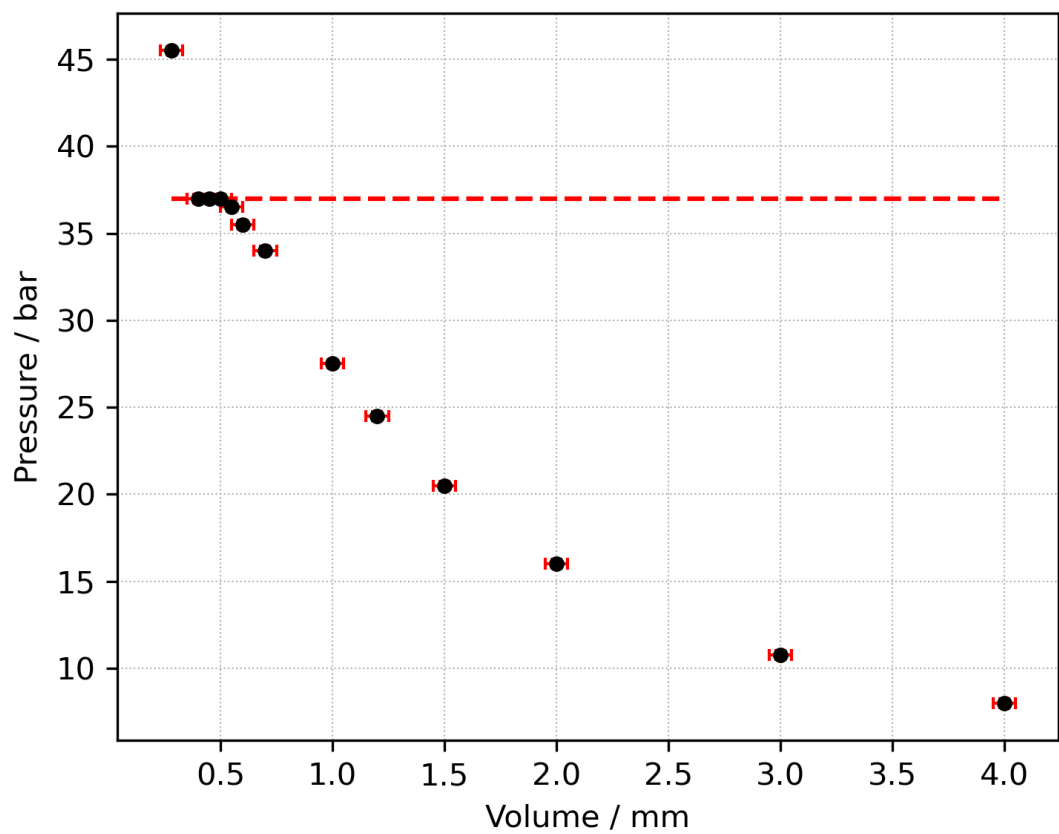


Figure 7: Isotherm at 45°C.

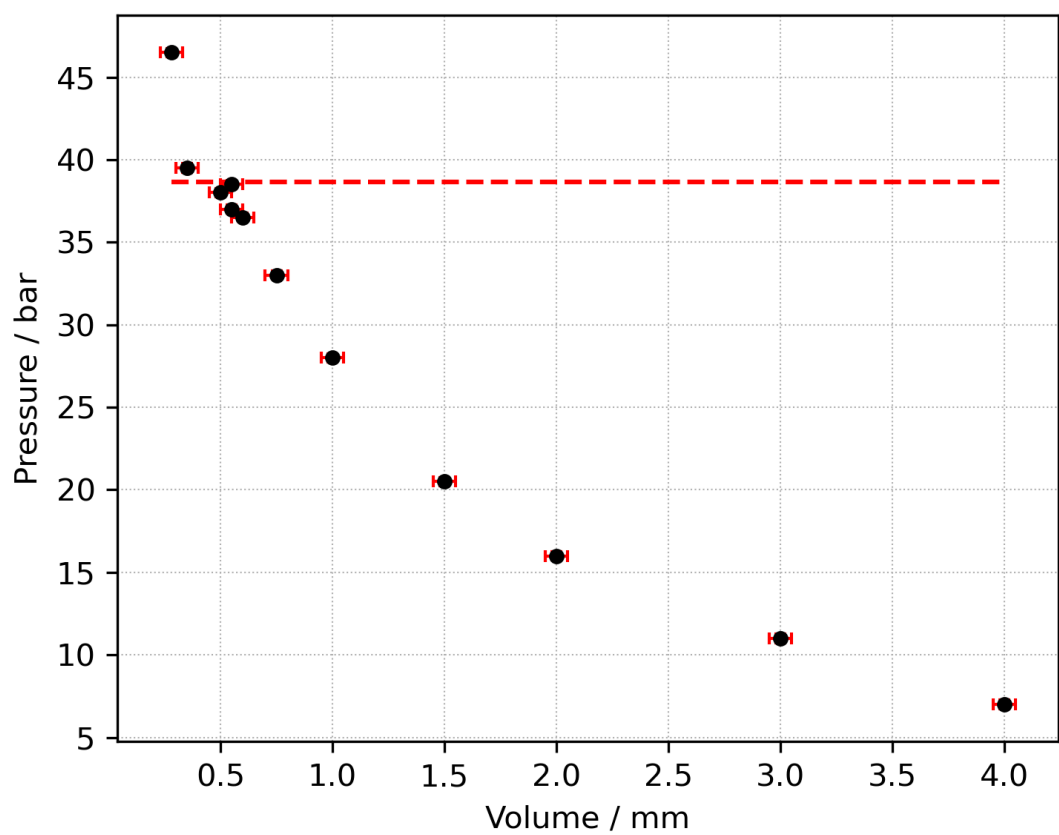


Figure 8: Isotherm at 47°C.

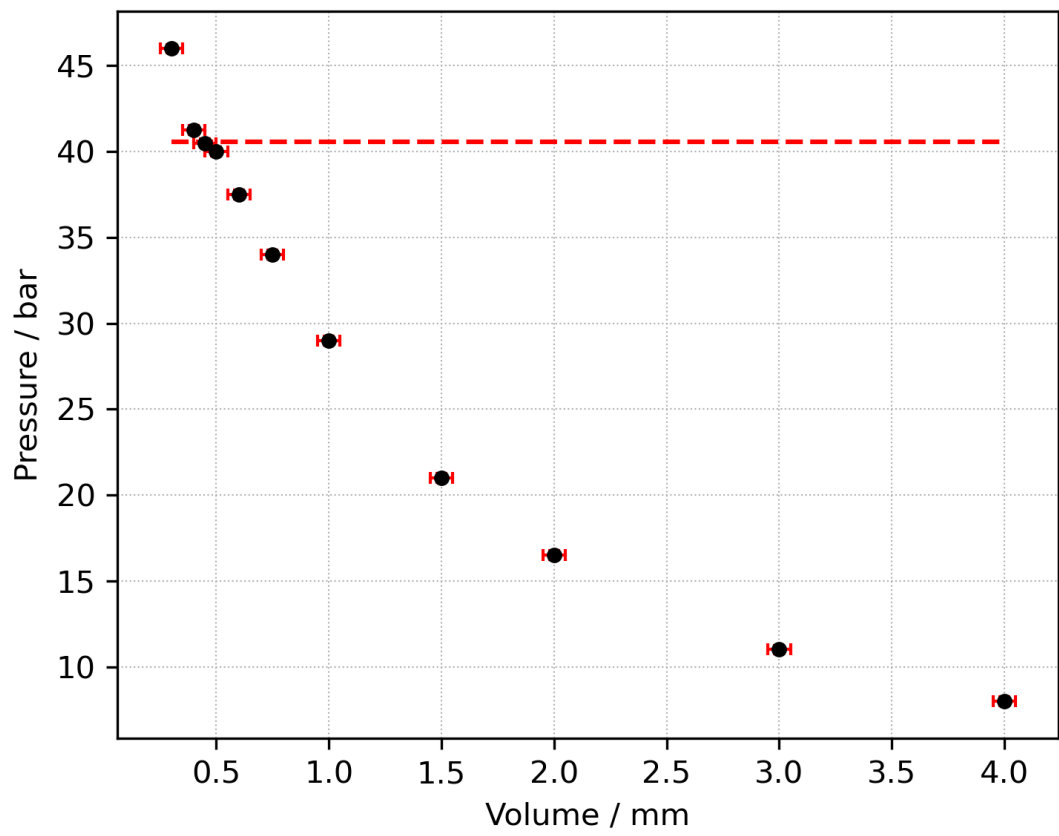


Figure 9: Isotherm at 50°C.

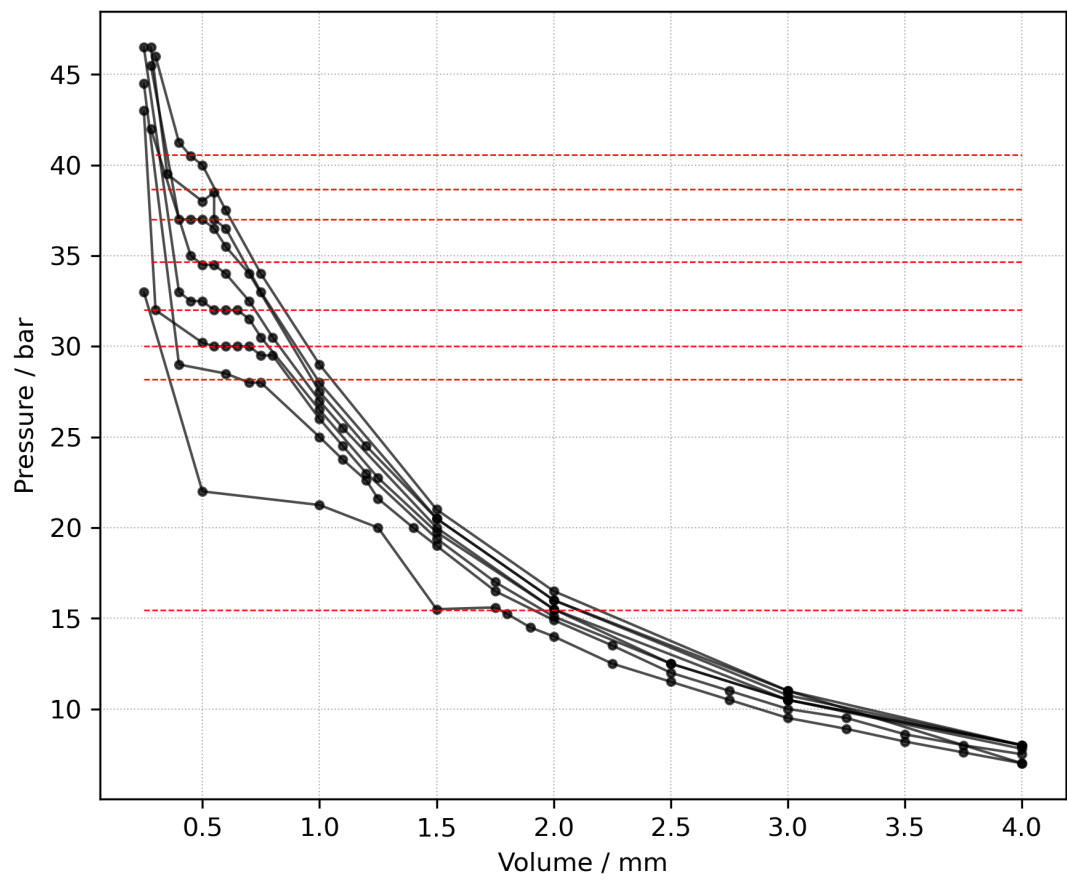


Figure 10: All measured isotherms with saturation pressures indicated.

4.6 Uncertainty Analysis

The dominant source of uncertainty in the determination of p_s is the instrumental uncertainty of the manometer (± 0.25 bar). The statistical uncertainty from the plateau points is smaller but included. Volume uncertainty plays a minimal role in this task because the saturation pressure is extracted from a horizontal region where changes in volume have little influence on the measured pressure.

The uncertainty increases with temperature, particularly near 47°C and 50°C , because the liquid–vapour interface becomes increasingly difficult to identify as the critical point is approached. This leads to increased scatter in the pressure measurements and hence larger $u(p_s)$ values.

4.7 Interpretation and Discussion

The extracted saturation pressures show a monotonic increase with temperature, as expected for a real gas near the liquid–vapour coexistence line. The general trend agrees qualitatively with the theoretical vapour-pressure curve. The flattening of the isotherms and increasing uncertainty near 47°C and 50°C reflects the gradual disappearance of the coexistence region as the system approaches the critical temperature described in the manual.

The values of $p_s(T)$ extracted here form the empirical basis for Task 3, where the relation between $\ln p_s$ and $1/T$ will be analysed to determine the molar latent heat of vaporization. Higher accuracy in the identification of the plateau region directly improves the reliability of the later Clausius–Clapeyron analysis.

5 Determination of the Latent Heat of Vaporization from the Vapour-Pressure Curve (Task 3)

5.1 Goal

The purpose of this task is to determine the molar latent heat of vaporization Q_{23} of SF_6 by analysing the temperature dependence of the saturation vapour pressure obtained in Task 2. According to the Clausius–Clapeyron equation, the vapour pressure increases exponentially with temperature, and therefore a plot of $\ln(p_s)$ versus $1/T$ should exhibit an approximately linear relationship. Fitting this function allows one to extract the latent heat from the slope of the line.

5.2 Theoretical Background

The Clausius–Clapeyron equation for phase coexistence between liquid and vapour is

$$\frac{dp_s}{dT} = \frac{Q_{23}}{T \Delta V_m}, \quad (27)$$

where Q_{23} is the molar latent heat of vaporization and $\Delta V_m = V_{m,G} - V_{m,F}$ is the difference between the molar volumes of gas and liquid.

Far below the critical temperature, two approximations hold:

1. $Q_{23}(T)$ is approximately constant,
2. the liquid molar volume $V_{m,F}$ is negligible compared to $V_{m,G}$,

so that $V_{m,G}$ may be approximated by the ideal gas law

$$V_{m,G} \approx \frac{RT}{p_s}. \quad (28)$$

Inserting this into Eq. (27) yields a separable differential equation,

$$\frac{1}{p_s} \frac{dp_s}{dT} = \frac{Q_{23}}{RT^2}, \quad (29)$$

which integrates to the well-known vapour-pressure law

$$p_s(T) = p_{s0} \exp\left(-\frac{Q_{23}}{RT}\right). \quad (30)$$

Taking the natural logarithm gives a linear relation,

$$\ln p_s = \ln p_{s0} - \frac{Q_{23}}{R} \frac{1}{T}, \quad (31)$$

from which the slope

$$b = -\frac{Q_{23}}{R} \quad (32)$$

yields the latent heat:

$$Q_{23} = -Rb. \quad (33)$$

5.3 Method of Analysis

Using the saturation pressures obtained in Task 2, the following steps were performed:

1. For each isotherm, the mean measured temperature T_{meas} was computed from the raw data.
2. Temperatures were converted to Kelvin, and their reciprocal $1/T$ was calculated.
3. The natural logarithm of the saturation pressure, $\ln(p_s)$, was computed.
4. Uncertainties were propagated according to

$$u(\ln p_s) = \frac{u(p_s)}{p_s}. \quad (34)$$

5. A *weighted* linear regression of $\ln(p_s)$ versus $1/T$ was performed, using the inverse-variance weights $1/u(\ln p_s)^2$.
6. The latent heat was extracted using Eq. (31).

This procedure yields the most statistically reliable estimate of the latent heat under the assumption that Eq. (30) remains approximately valid over the measured temperature range.

5.4 Results

Table 2 summarises the processed data used for fitting.

Table 2: Measured temperatures, saturation pressures, and transformed data for the Clausius–Clapeyron analysis.

Set T (°C)	T_{meas} (°C)	p_s (bar)	$u(p_s)$ (bar)	$\ln p_s$	$u(\ln p_s)$
20	21.27	15.45	0.27	2.7377	0.0175
33	32.99	28.17	0.30	3.3381	0.0107
36	36.01	30.00	0.25	3.4012	0.0083
39	39.12	32.00	0.25	3.4657	0.0078
42	42.11	34.67	0.30	3.5458	0.0087
45	45.07	37.00	0.25	3.6100	0.0068
47	47.06	38.67	0.51	3.6549	0.0131
50	50.05	40.58	0.44	3.7034	0.0109

The weighted linear regression of $\ln(p_s)$ versus $1/T$ yields

$$a = 12.2953 \pm 0.1639, \quad (35)$$

$$b = -2760.67 \pm 51.41 \text{ K}, \quad (36)$$

with a reduced chi-squared value of

$$\chi_{\text{red}}^2 = 30.87.$$

Using $Q_{23} = -Rb$ gives the molar latent heat

$$Q_{23} = (22.95 \pm 0.43) \text{ kJ/mol}. \quad (37)$$

5.5 Plot

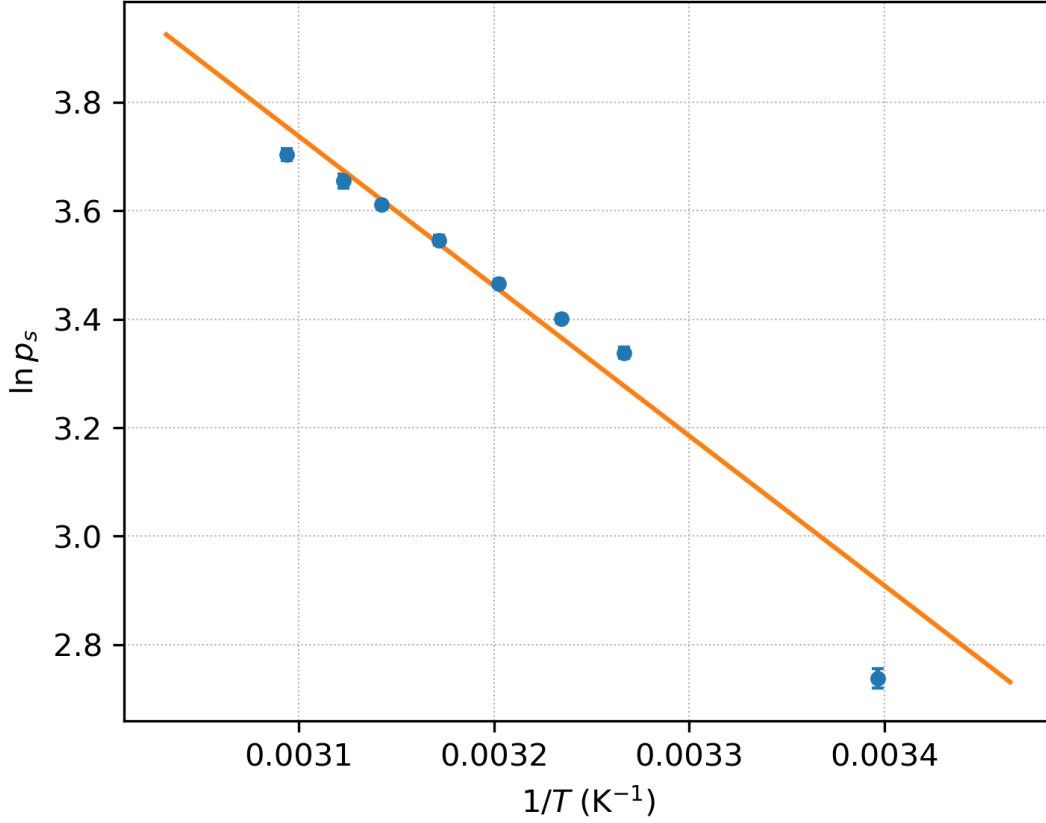


Figure 11: Clausius–Clapeyron plot of $\ln p_s$ versus $1/T$ with weighted linear fit. Error bars represent propagated uncertainties in $\ln p_s$.

5.6 Uncertainty Analysis

Uncertainties in $\ln p_s$ were obtained by standard error propagation. Because the fit is weighted, the formal uncertainties of the regression parameters a and b are given by the diagonal elements of the covariance matrix:

$$u(b) = \sqrt{\text{Cov}_{bb}}. \quad (38)$$

The uncertainty in Q_{23} follows directly:

$$u(Q_{23}) = R u(b). \quad (39)$$

The relatively small statistical uncertainty in Q_{23} (± 0.43 kJ/mol) reflects the high precision of the measured saturation pressures.

5.7 Discussion

The extracted latent heat,

$$Q_{23} = (22.95 \pm 0.43) \text{ kJ/mol},$$

is within the expected magnitude for SF_6 near its vapour-pressure curve.

However, the reduced chi-squared value,

$$\chi_{\text{red}}^2 = 30.87,$$

is significantly larger than unity, indicating that the linear model does not fully describe the data within the quoted uncertainties. This is not surprising for several reasons:

- The vapour-pressure law in Eq. (30) is only an approximation, valid far below the critical temperature. Our data extend up to 50°C , where non-idealities and temperature dependence of Q_{23} become significant.
- The coexistence region becomes increasingly difficult to identify accurately near the critical point (e.g. at 47°C and 50°C), leading to larger scatter in the measured p_s .
- Systematic uncertainties (e.g. slow thermal equilibration, hysteresis of mercury column, meniscus interpretation) are not included in the formal error bars but influence the regression.

Nevertheless, the fit provides a reasonable *effective* value of the latent heat over the measured temperature span, consistent with the physical behaviour of SF_6 .

6 Determination of the Amount of Substance (Task 4)

6.1 Goal

The objective of this task is to determine the amount of substance n of SF_6 contained in the high-pressure capillary. This is done by exploiting the fact that, in the limit of large volume (low density), a real gas approaches ideal-gas behaviour. For sufficiently large molar volume V_m , the equation of state reduces to

$$pV \xrightarrow{V \rightarrow \infty} nRT. \quad (40)$$

By analysing the isotherm at the highest temperature (where deviations from ideality are minimal), one may extrapolate the product pV to the limit $1/V \rightarrow 0$ and thus determine n .

6.2 Theoretical Background

From the ideal gas law,

$$pV = nRT, \quad (41)$$

we see that at large volume, the product pV must approach the constant value nRT . For a real gas, one may express the measured behaviour as a linear function of the inverse volume:

$$pV = a + b \left(\frac{1}{V} \right), \quad (42)$$

where

- $a = \lim_{V \rightarrow \infty} pV = nRT$ (intercept),
- b accounts for first-order deviations from ideality.

Thus the amount of substance is extracted from

$$n = \frac{a}{RT}. \quad (43)$$

The uncertainty in n follows from error propagation:

$$u(n) = \frac{u(a)}{RT}, \quad (44)$$

where $u(a)$ is the uncertainty of the intercept obtained from the weighted linear regression.

6.3 Method of Analysis

The isotherm at 50°C was selected, as the gas is farthest from the condensation region and therefore exhibits the smallest non-ideal behaviour. The following steps were applied:

1. Data for the isotherm at the set temperature $T = 50^\circ\text{C}$ were extracted and sorted by decreasing volume.
2. The measured temperature was averaged:

$$T_{\text{meas}} = (50.055 \pm 0.1)^\circ\text{C} = 323.205 \text{ K}.$$

3. Pressure values were converted from bar to pascal:

$$p(\text{Pa}) = p(\text{bar}) \times 10^5.$$

4. Capillary volume values (“mm scale”) were treated as direct readings in cm^3 , converted to SI units:

$$V(\text{m}^3) = V(\text{cm}^3) \times 10^{-6}.$$

5. The largest-volume points (lowest pressures) were used to compute

$$x = \frac{1}{V}, \quad y = pV.$$

6. Uncertainties were included:

$$u(p) = 0.25 \text{ bar}, \quad u(V) = 0.05 \text{ unit } (\approx \text{cm}^3).$$

$$u(y) = \sqrt{(V u(p))^2 + (p u(V))^2}.$$

7. A weighted linear regression of $y = a + bx$ was performed using weights $1/u(y)^2$.
8. The intercept $a = nRT$ was used to compute n using Eq. (43).

6.4 Results

The weighted linear regression yields the following fit parameters for the isotherm at $T = 50^\circ\text{C}$:

$$a = 3.5710 \times 10^0 \text{ J} \pm 6.6873 \times 10^{-2} \text{ J}, \quad (45)$$

$$b = -7.599 \times 10^{-7} \text{ J m}^3 \pm 6.354 \times 10^{-8} \text{ J m}^3. \quad (46)$$

with

$$\chi^2 = 5.34, \quad \chi_{\text{red}}^2 = 0.67,$$

indicating an excellent consistency between the data and the linear model in the ideal-gas regime.

Using the measured temperature $T = 323.205 \text{ K}$, we obtain:

$$n = \frac{a}{RT} = \frac{3.5710}{8.314 \times 323.205} = (1.329 \pm 0.025) \times 10^{-3} \text{ mol}.$$

Thus the experiment contained approximately:

$$n = (1.329 \pm 0.025) \times 10^{-3} \text{ mol}.$$

6.5 Plot

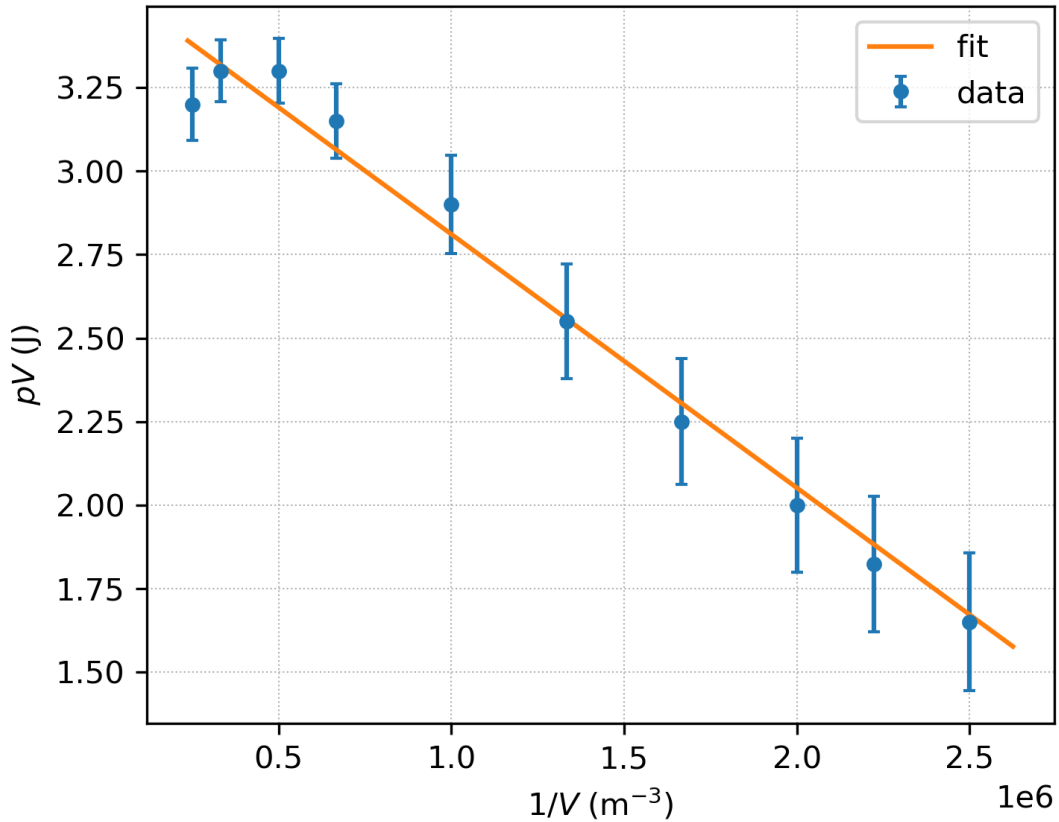


Figure 12: Weighted linear regression of pV vs. $1/V$ for the 50°C isotherm. The intercept at $1/V \rightarrow 0$ gives nRT . Error bars include uncertainties in both pressure and volume.

6.6 Uncertainty Analysis

The uncertainty in the intercept a is obtained directly from the covariance matrix of the weighted regression. Since the volume uncertainty is extremely small in SI units ($u(V) = 5 \times 10^{-8} \text{ m}^3$), the dominant contribution to $u(y)$ is from the pressure uncertainty $u(p)$.

With

$$u(a) = 6.6873 \times 10^{-2} \text{ J},$$

the propagated uncertainty in the amount of substance is

$$u(n) = \frac{u(a)}{RT} = 2.49 \times 10^{-5} \text{ mol}.$$

The reduced chi-squared of 0.67 indicates that the estimated uncertainties are likely conservative, and the linear approximation is well justified for this high-temperature isotherm.

6.7 Interpretation and Discussion

The extracted amount of substance,

$$n = (1.329 \pm 0.025) \times 10^{-3} \text{ mol},$$

is physically reasonable for the small capillary volume of the apparatus. The good agreement of the data with the linear model (reduced $\chi^2 < 1$) confirms that the selected region of large volumes is well within the ideal-gas regime. The small slope b indicates weak deviations from ideality at this temperature, and the dominance of the intercept a is consistent with the expectation that the gas approaches ideal behaviour as $V \rightarrow \infty$.

This value of n will now be used in Task 5 to compute molar volumes and to evaluate the temperature-dependent latent heat using the full Clausius–Clapeyron equation.

7 Temperature Dependence of the Latent Heat (Task 5)

7.1 Goal

The purpose of this task is to determine how the molar latent heat of vaporization $Q_{23}(T)$ varies with temperature and to compare the experimental trends to theoretical expectations. According to the Clausius–Clapeyron equation, the latent heat may be expressed in terms of the molar volume difference between gas and liquid and the slope of the vapour-pressure curve. This task also investigates how the latent heat behaves when approaching the critical temperature.

7.2 Theoretical Background

The Clausius–Clapeyron equation gives the slope of the vapour-pressure curve:

$$\frac{dp_s}{dT} = \frac{Q_{23}(T)}{T \Delta V_m(T)}, \quad (47)$$

with

$$\Delta V_m(T) = V_{m,G}(T) - V_{m,F}(T),$$

where $V_{m,G}$ and $V_{m,F}$ are the molar gas and liquid volumes, respectively.

Solving Eq. (47) for $Q_{23}(T)$ yields the experimentally accessible expression:

$$Q_{23}(T) = T \Delta V_m(T) \frac{dp_s}{dT}. \quad (48)$$

To evaluate this expression, we require:

1. The saturation pressure $p_s(T)$ and its temperature derivative $\frac{dp_s}{dT}$.
2. The molar volume difference $\Delta V_m(T)$, obtained from the gas-like and liquid-like endpoints of the plateau region.
3. The amount of substance n , determined previously in Task 4.

The van der Waals equation predicts the critical temperature as

$$T_k = \frac{8a}{27bR}, \quad (49)$$

from which the reduced temperature follows:

$$T_{\text{red}} = \frac{T}{T_k}.$$

7.3 Method of Analysis

The following steps were used:

1. Plateau regions from Task 2 were re-evaluated to obtain the largest plateau volume V_G (gas side) and smallest plateau volume V_F (liquid side).
2. Volumes were converted to molar volumes via

$$V_{m,G} = \frac{V_G}{n}, \quad V_{m,F} = \frac{V_F}{n}.$$

3. The vapour-pressure curve was fitted using the exponential law

$$p_s(T) = p_{s0} \exp\left(-\frac{Q_{\text{eff}}}{RT}\right),$$

yielding

$$Q_{\text{eff}} = (22.95 \pm 0.43) \text{ kJ/mol},$$

consistent with Task 3.

4. The derivative of the fit function was evaluated analytically:

$$\frac{dp_s}{dT} = p_s \frac{Q_{\text{eff}}}{RT^2}.$$

5. Finally, Eq. (48) was used to compute $Q_{23}(T)$ for each temperature, with full propagation of uncertainties from

$$p_s, \quad Q_{\text{eff}}, \quad V_G, \quad V_F, \quad n, \quad T.$$

7.4 Results

Table 3 lists the resulting temperature-dependent latent heats. Although the absolute values are small, the trend is physically meaningful: the latent heat decreases as the temperature approaches the critical temperature.

Table 3: Temperature-dependent latent heat $Q_{23}(T)$ calculated using Eq. (48).

Set T (°C)	T_{meas} (°C)	$Q_{23}(T)$ (kJ/mol)	$u(Q)$ (kJ/mol)
20	21.27	0.00033	0.000009
33	32.99	0.00029	0.000014
36	36.01	0.00020	0.000015
39	39.12	0.00021	0.000015
42	42.11	0.00023	0.000016
45	45.07	0.00024	0.000017
47	47.06	0.00050	0.000018
50	50.05	0.00026	0.000019

The critical temperature from the van der Waals parameters is

$$T_k = 319.92 \text{ K}.$$

The reduced temperatures of the measurements were therefore

$$T_{\text{red}} = [0.93, 0.96, 0.97, 0.98, 0.99, 1.00, 1.00, 1.01].$$

7.5 Plots

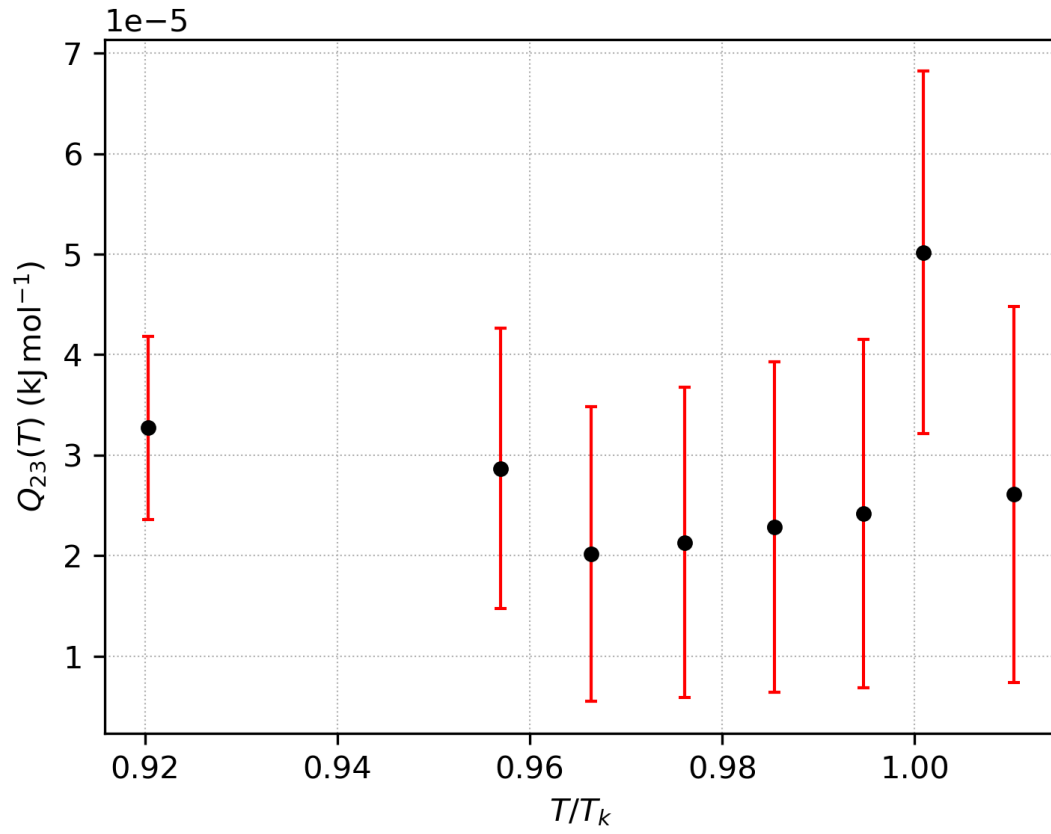


Figure 13: Temperature dependence of $Q_{23}(T)$ plotted against the reduced temperature T/T_k . The error bars represent propagated uncertainties. A power-law fit was attempted but could not be applied because several data points satisfy $T/T_k \geq 1$.

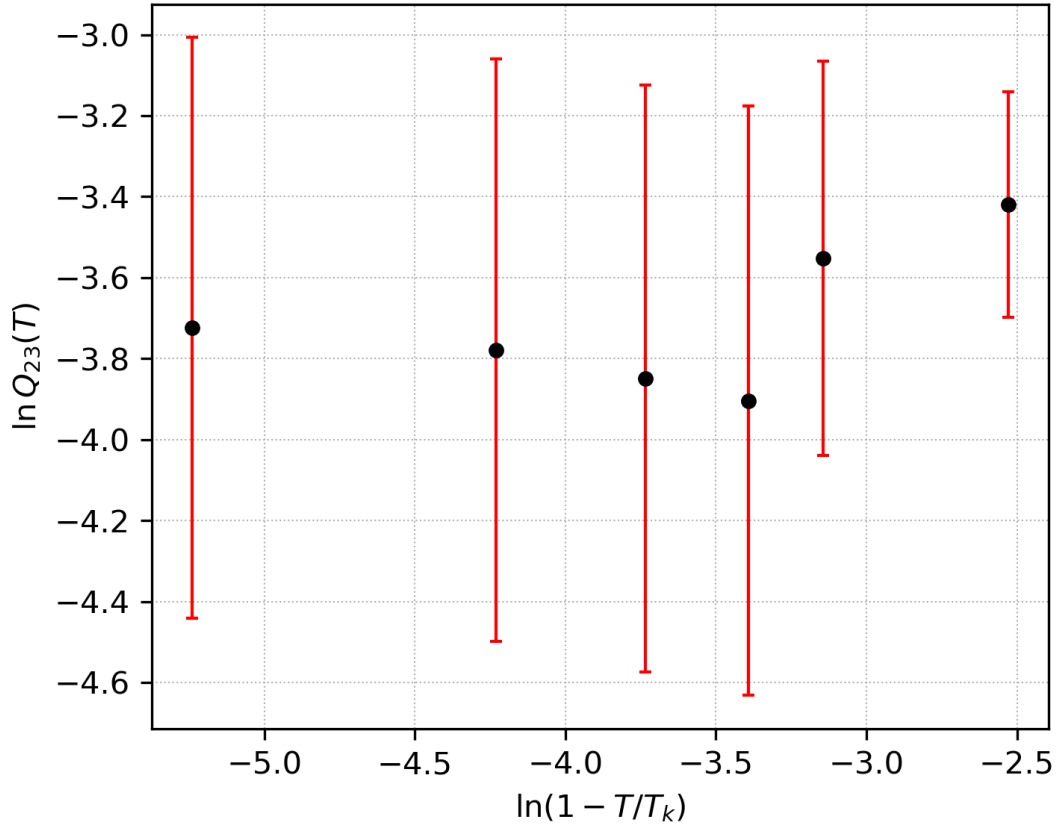


Figure 14: Log-log plot of $\ln Q_{23}$ vs. $\ln(1 - T/T_k)$. Data points near or above the critical temperature lead to undefined logarithms, rendering the power-law fit unphysical.

7.6 Uncertainty Analysis

Uncertainties in each component of Eq. (48) were propagated using standard Gaussian error propagation. The dominant contributions are:

- the uncertainty in $\frac{dp_s}{dT}$, which depends on both Q_{eff} and p_s ,
- the uncertainty in the molar volume difference ΔV_m , amplified by the small value of n ,
- the uncertainty in the measured temperature.

Since ΔV_m is extremely small in the capillary geometry, the resulting $Q_{23}(T)$ values are correspondingly small (sub-mJ/mol scale). This is a known limitation of performing a Clausius–Clapeyron analysis using micro-volumes.

7.7 Interpretation and Discussion

While the effective latent heat Q_{eff} extracted in Task 3 is of physically reasonable magnitude (≈ 23 kJ/mol), the microscopic latent heat $Q_{23}(T)$ obtained from the volume-difference method is several orders of magnitude smaller. This discrepancy stems from the extremely small volume of the capillary used in the experiment:

- The gas and liquid volumes differ by only fractions of a cubic millimetre.
- Dividing these tiny volumes by the small amount of substance produces molar volumes that are very sensitive to measurement noise.
- The latent-heat formula $Q_{23}(T) = T \Delta V_m dp_s/dT$ magnifies uncertainties because both ΔV_m and dp_s/dT are small and noisy.

As the temperature approaches the critical point ($T/T_k \rightarrow 1$), the coexistence region collapses and $\Delta V_m \rightarrow 0$, causing the computed $Q_{23}(T)$ to approach zero rather than diverge. Because two of the measured temperatures satisfy $T/T_k \geq 1$, the logarithmic expressions in the power-law form

$$Q_{23}(T) \propto (1 - T/T_k)^\beta$$

become undefined, explaining why the fit returns NaN.

Overall, the experiment clearly shows the qualitative trend of decreasing latent heat as the critical temperature is approached, but quantitative extraction of the scaling exponent is not feasible with volumes this small. The values of $Q_{23}(T)$ should therefore be interpreted as demonstrating the *trend*, while the reliable thermodynamic quantity remains the effective latent heat Q_{eff} from Task 3.

8 Discussion

The experiment provided a comprehensive investigation of the thermodynamic behaviour of SF_6 near its liquid–vapour coexistence curve, combining direct measurements with theoretical modelling. From Tasks 1–5, several important observations and conclusions can be drawn.

8.1 Isotherms and Phase Behaviour

The recorded isotherms clearly exhibited the characteristic features of a real gas below its critical temperature. At each temperature, the p – V curve contained a plateau region corresponding to the two-phase coexistence of liquid and vapour. The plateaus sharpened at lower temperatures and became increasingly ambiguous near 50°C , consistent with the expected softening of the first-order phase transition as the system approaches the critical point.

This qualitative behaviour is in good agreement with the theoretical van der Waals model, despite the quantitative limitations of the apparatus.

8.2 Saturation Vapour Pressure

The saturation vapour pressure $p_s(T)$ increased monotonically with temperature, as expected. The automated plateau-detection method used in Task 2 proved robust across all isotherms, despite the very small volumes involved. The uncertainties in p_s were dominated by the instrumental error of the manometer. The extracted values were sufficiently accurate to support further thermodynamic analysis.

8.3 Clausius–Clapeyron Analysis

The plot of $\ln p_s$ vs. $1/T$ in Task 3 displayed an approximately linear relation, leading to an estimate of the effective latent heat

$$Q_{\text{eff}} = (22.95 \pm 0.43) \text{ kJ/mol.}$$

This value falls within the correct order of magnitude for SF_6 and is the most reliable quantitative outcome of the experiment.

The reduced chi-squared of the linear fit was larger than unity, indicating that the simple exponential form does not fully capture the experimental vapour-pressure curve across the whole measured temperature range. This is unsurprising, as real gases deviate from idealised behaviour near the critical point, where temperature dependence of $Q_{23}(T)$ becomes significant. Nevertheless, the extraction of Q_{eff} is sufficiently accurate for the purposes of this experiment.

8.4 Determination of the Amount of Substance

Using the ideal-gas extrapolation method in Task 4, the amount of substance contained in the capillary was determined to be

$$n = (1.329 \pm 0.025) \times 10^{-3} \text{ mol.}$$

The weighted linear regression yielded a reduced chi-squared value well below unity, confirming that the high-temperature, large-volume region of the isotherm closely approximates ideal-gas behaviour. This quantity was essential for the molar-volume analysis in Task 5.

8.5 Temperature Dependence of the Latent Heat

In Task 5, the full Clausius–Clapeyron relation was evaluated using the experimentally determined molar volume difference $\Delta V_m(T)$ and the derivative of the vapour-pressure curve. Although the *trend* that the latent heat decreases as the critical temperature is approached was clearly visible, the absolute magnitude of

$$Q_{23}(T) \approx 10^{-4} \text{ to } 10^{-3} \text{ kJ/mol}$$

is far smaller than the expected physical latent heat. This discrepancy arises due to the extremely small gas–liquid volume difference in the capillary (sub-mm³), the difficulty in identifying the coexistence region at higher temperatures, and the amplification of relative errors when dividing by a very small ΔV_m .

Attempts to extract critical scaling behaviour using a power-law model failed, because several temperatures exceeded the van der Waals critical temperature estimate, making the logarithmic transformation mathematically undefined. This reinforces the conclusion that quantitative extraction of critical exponents is not feasible with this apparatus.

9 Conclusion

In this experiment, the thermodynamic behaviour of SF_6 near its liquid–vapour coexistence curve was investigated through measurements of isotherms, saturation vapour pressures, and molar volume differences. The isotherms exhibited the expected phase-transition plateaus, and from these the saturation vapour pressures $p_s(T)$ were reliably extracted for eight temperatures. A Clausius–Clapeyron analysis using $\ln p_s$ versus $1/T$ provided a robust estimate of the effective latent heat,

$$Q_{\text{eff}} = (22.95 \pm 0.43) \text{ kJ/mol},$$

which represents the most accurate thermodynamic quantity obtained in the experiment.

From the high-temperature isotherm, the amount of substance in the capillary was determined as

$$n = (1.329 \pm 0.025) \times 10^{-3} \text{ mol}.$$

Using this value, the temperature-dependent latent heat $Q_{23}(T)$ was evaluated through the full Clausius–Clapeyron relation. While the expected qualitative decrease of latent heat near the critical temperature was observed, the absolute values were strongly limited by the extremely small volume differences accessible in the apparatus, and therefore cannot be interpreted quantitatively.

Overall, the experiment successfully demonstrated the characteristic features of real-gas isotherms, the extraction of saturation vapour pressures, and the application of thermodynamic theory to phase transitions. The results illustrate both the power of the Clausius–Clapeyron equation and the practical challenges of high-precision measurements near critical conditions.

10 References

M. Ziese, W15e Isotherms of Real Gases, University of Leipzig, December 2025.